

Decent Coaching Classes

VIII Sarvadnya

Worksheet

Factorization

Topic: Common Factors of Monomials & Polynomials

Marks: 30

Section A — HCF of Monomials ($10 \times 1 = 10$ Marks)

Find the Highest Common Factor (HCF) of the following monomials.

1. Find the HCF of: $12x^2y$ and $18xy^2$ [1]
2. Find the HCF of: $15a^3b^2$ and $25a^2b^3$ [1]
3. Find the HCF of: $8m^2n$, $12mn^2$ and $20m^2n^2$ [1]
4. Find the HCF of: $9p^3q^2r$ and $27p^2q^3r^2$ [1]
5. Find the HCF of: $14x^2yz$, $21xy^2z$ and $35xyz^2$ [1]
6. Find the HCF of: $16a^4b^3$ and $24a^3b^4$ [1]
7. Find the HCF of: $5x$, $10x^2$ and $15x^3$ [1]
8. Find the HCF of: $18m^3n^2p$ and $24m^2n^3p^2$ [1]
9. Find the HCF of: $20a^2bc$, $30ab^2c$ and $50abc^2$ [1]
10. Find the HCF of: $36x^3y^4$, $48x^4y^3$ and $60x^2y^2$ [1]

Section B — Common Factorization of Polynomials ($10 \times 2 = 20$ Marks)

Factorize the following polynomials by taking out the common factor.

1. Factorize: $5x + 15$ [2]
2. Factorize: $7a^2 - 14a$ [2]
3. Factorize: $9x^2y + 12xy^2$ [2]
4. Factorize: $3m^3 - 6m^2 + 9m$ [2]
5. Factorize: $4p^2q - 8pq^2 + 12pq$ [2]
6. Factorize: $6a^2b - 9ab^2 + 12ab$ [2]
7. Factorize: $15x^3y^2 - 20x^2y^3 + 25x^2y^2$ [2]
8. Factorize: $8m^2n^2 + 12mn^3 - 16m^2n$ [2]
9. Factorize: $5(x + 2) + 10y(x + 2)$ [2]
10. Factorize: $3a(x - y) - 6b(x - y)$ [2]

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Answer Key & Solutions

Section A — HCF of Monomials

1. $\text{HCF} = 6xy$ (coefficients: HCF of 12, 18 = 6; common letters with lowest powers: x^1, y^1)
2. $\text{HCF} = 5a^2b^2$ (HCF of 15, 25 = 5; lowest powers: a^2, b^2)
3. $\text{HCF} = 4mn$ (HCF of 8, 12, 20 = 4; lowest powers: m^1, n^1)
4. $\text{HCF} = 9p^2q^2r$ (HCF of 9, 27 = 9; lowest powers: p^2, q^2, r^1)
5. $\text{HCF} = 7xyz$ (HCF of 14, 21, 35 = 7; lowest powers: x^1, y^1, z^1)
6. $\text{HCF} = 8a^3b^3$ (HCF of 16, 24 = 8; lowest powers: a^3, b^3)
7. $\text{HCF} = 5x$ (HCF of 5, 10, 15 = 5; lowest power: x^1)
8. $\text{HCF} = 6m^2n^2p$ (HCF of 18, 24 = 6; lowest powers: m^2, n^2, p^1)
9. $\text{HCF} = 10abc$ (HCF of 20, 30, 50 = 10; lowest powers: a^1, b^1, c^1)
10. $\text{HCF} = 12x^2y^2$ (HCF of 36, 48, 60 = 12; lowest powers: x^2, y^2)

Section B — Common Factorization of Polynomials

1. $5x + 15 = 5(x + 3)$
2. $7a^2 - 14a = 7a(a - 2)$
3. $9x^2y + 12xy^2 = 3xy(3x + 4y)$
4. $3m^3 - 6m^2 + 9m = 3m(m^2 - 2m + 3)$
5. $4p^2q - 8pq^2 + 12pq = 4pq(p - 2q + 3)$
6. $6a^2b - 9ab^2 + 12ab = 3ab(2a - 3b + 4)$
7. $15x^3y^2 - 20x^2y^3 + 25x^2y^2 = 5x^2y^2(3x - 4y + 5)$
8. $8m^2n^2 + 12mn^3 - 16m^2n = 4mn(2mn + 3n^2 - 4m)$
9. $5(x + 2) + 10y(x + 2) = 5(x + 2)(1 + 2y)$
10. $3a(x - y) - 6b(x - y) = 3(x - y)(a - 2b)$